

Unit-5 Algebraic Structure

Binary Operation $\Rightarrow$ An operation which combines two elements of a set to give another element of the same set is called the binary operation.

Mathematically, let A be a non-empty set then $A \times A = \{(a, b) : a \in A, b \in A\}$. If f is any mapping from $A \times A \rightarrow A$ i.e. $f: A \times A \rightarrow A$, then f is called binary operation on A .

OK A binary operation $*$ on a set G is a function $f: A \times G \times G \rightarrow G$ which associates every ordered pair of elements x, y to G some unique element $x * y$ of G .

NOTE $\Rightarrow$ " $*$ " will be a binary operation on set A , if $a \in A, b \in A \Rightarrow a * b \in A$ and $a * b$ is unique.

Example $\Rightarrow$ $A = \mathbb{N} = \{1, 2, 3, \dots\}$
 $+ : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$

Let $a, b \in \mathbb{N}$ then $(a, b) \in \mathbb{N} \times \mathbb{N}$
 $\Rightarrow (a, b) \rightarrow a + b = c$ (unique)

$2, 3 \in \mathbb{N}$ then $(2, 3) \rightarrow 2 + 3 = 5 \in \mathbb{N}$

NOTE $\Rightarrow$

① Addition is a binary operation on set N .

Since $a, b \in N \Rightarrow (a+b) \in N, \forall a, b \in N$.

② Subtraction is not a binary operation on the set N .

③ Subtraction is binary operation on the set Z

of integers as $a, b \in Z \Rightarrow (a-b) \in Z, \forall a, b \in Z$.

Properties $\Rightarrow$

① Commutative $\Rightarrow$ A binary operation $*$ on a set A is called commutative if $a * b = b * a, \forall a, b \in A$.

Ex $\Rightarrow$ Binary operation of addition on set of integers is commutative. $a+b = b+a$

Binary operation of subtraction on set of integers is not commutative. $a-b \neq b-a$

② Associative $\Rightarrow$ A binary operation $*$ on set A is associative if $(a * b) * c = a * (b * c), \forall a, b, c \in A$.

Ex $\Rightarrow$ Binary operation of multiplication on Z is associative as $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

Binary operation of subtraction on Z is not associative $(2-3)-4 \neq 2-(3-4)$

Algebraic structure $\Rightarrow$

A structure consisting of a set and one or more binary operations on the set is called Algebraic structure.

Represented as $\Rightarrow (A, f_1, f_2, \dots)$

$A \rightarrow$ non-empty set

$f_1, f_2 \rightarrow$ Binary operation on A .

Ex $\Rightarrow (N, +) \Rightarrow$ Algebraic structure

$(R, +, \cdot) \Rightarrow$ Algebraic structure.

$\uparrow$
mul.

Properties $\Rightarrow$

① Closure property $\Rightarrow$ An operation "*" is called on a set A is called closure, if

$$a, b \in A \Rightarrow a * b \in A \quad \forall a, b \in A$$

Ex $\Rightarrow (N, +) \rightarrow$ Closure $(N, -) \rightarrow$ Not Closure

$(N, \cdot) \rightarrow$ Closure

② Associative property $\Rightarrow$ An operation "*" is on a

set A is called associative, if

$$a * (b * c) = a * (b * c) \quad \forall a, b, c \in A.$$

Ex $\Rightarrow (Z, \cdot) \rightarrow$ associative
 $\uparrow$ mul

$(Z, +) \rightarrow$ Associative
 $(Z, -) \rightarrow$ Not associative

③ Existence of Identity $\Rightarrow$

An element "e" is an algebraic structure $(A, *)$ is called an identity element, if

$$e * a = a = a * e \quad \forall a \in A.$$

Eg $\Rightarrow$ ① "0" is an identity element in $(\mathbb{Z}, +)$

$$0 + a = a = a + 0 \quad \forall a \in \mathbb{Z}$$

② "1" is an identity element in $(\mathbb{N}, \cdot)$

$$1 \cdot a = a = a \cdot 1 \quad \forall a \in \mathbb{N}$$

④ Existence of Inverse $\Rightarrow$ for every $a \in A$, $b \in A$

such that $a * b = e = b * a$, where "e" is an identity element, then

The element b is called the inverse of a and denoted by a^{-1} .

Eg $\Rightarrow$ ① $2 + (-2) = 0 = (-2) + 2 = 0$

② $\frac{1}{3} \times 3 = 1 = 3 \times \frac{1}{3}$

⑤ Commutative Property $\Rightarrow$ An operation "*" on a set

A is called Commutative if

$$a * b = b * a \quad \forall a, b \in A$$

Eg. $\Rightarrow N = \{1, 2, 3, 4, \dots\}$

$\Rightarrow (N, +) \rightarrow$ Commutative.

& $(N, -) \rightarrow$ Not Commutative.

Semi-group $\Rightarrow$ An algebraic structure $(A, *)$ is called a semigroup if the binary operation $*$ satisfies the closure and associative properties.

Semi-group $\Rightarrow$ Closure + Associative.

Ex $\Rightarrow$ ① $(\mathbb{N}, +) \Rightarrow$ Semi-group.

② $(\mathbb{Z}, -) \Rightarrow$ Semi-group.

Subsemigroup $\Rightarrow$ Let $(A, *)$ be a semi-group and B be a subset of A .

If B is closed under the operation $*$, the $(B, *)$ is called a subsemigroup.

Ex $\Rightarrow B = \{2, 4, 6, \dots\}$

then $(B, \cdot)$ is a subsemigroup of semigroup of $(\mathbb{Z}, \cdot)$.

Q: Prove that the intersection of two ^{Subsemi} semigroups of a semigroup $(A, *)$ is a subsemigroup of $(A, *)$

Sol: Let $(A_1, *)$ and $(A_2, *) \Rightarrow$ Subsemigroup.

To prove $\Rightarrow (A_1 \cap A_2, *) \Rightarrow$ Subsemigroup.

Cyclic Semigroup $\Rightarrow$ A semigroup $(A, *)$ is said to be cyclic if every element of A can be generated by some element a of A i.e. if $x \in A$ then $x = a * a * a * \dots * a$ (n times).
In this case, the element " a " is called the generator of A .

Q: Show that multiplicative group $\{1, -1, i, -i\}$ is cyclic.
And its generator. (RTU-2024)

Ans:

$$\begin{aligned} (i)^1 &= i & (-i)^1 &= -i \\ (i)^2 &= i^2 = -1 & (-i)^2 &= 1 \\ (i)^3 &= -i & (-i)^3 &= -1 \\ (i)^4 &= 1 & (-i)^4 &= 1 \end{aligned}$$

Generator $\Rightarrow \{i, -i\}$

monoid $\Rightarrow$ An algebraic structure $(A, *)$ is said to be a monoid, if it satisfies closure, Associative and existence of identity element properties.

monoid = closure + Associative + Identity element.

Eg $\Rightarrow (N, +) \rightarrow$ monoid with identity element 0.
 $(Z, +) \rightarrow$ ——— ——— ——— ——— ——— 0.

$(N, +) \rightarrow$ Not a monoid because

for addition 0 is the identity element but $0 \notin N$.

Submonoid $\Rightarrow$ Let $(A, *)$ be a monoid with identity "e" and let B be a non empty subset of A .

If B is closed under the operation $*$ and $e \in A$, then $(B, *)$ is called a submonoid.

Eg $\Rightarrow A = \text{set of even int} = \{ \dots, -4, -2, 0, 2, 4, \dots \}$

$(A, +) \rightarrow$ submonoid of $(Z, +)$

Cyclic monoid $\Rightarrow$ A monoid $(A, *)$ is said to be cyclic if every element of A can be generated by some element a of A i.e. if $x \in A$, then $a * a * a * \dots * a$ (n times).

The element " a " is called generator of A .

Q: Show that the multiplicative group $\{1, \omega, \omega^2\}$ is cyclic. Find its generator.

Sol:

$$(\omega)^1 = \omega$$

$$(\omega^2)^1 = \omega^2$$

$$(\omega)^2 = \omega^2$$

$$(\omega^2)^2 = \omega$$

$$(\omega)^3 = 1$$

$$(\omega^2)^3 = 1$$

$\vdots$

$\vdots$

Generator $\Rightarrow \{\omega, \omega^2\}$

Group $\Rightarrow$

$\Rightarrow$ An algebraic structure $(A, *)$ is called a group, if the binary operation $*$ satisfies the following conditions/property:

- (i) Closure property (ii) Associative property
(iii) Existence of identity (iv) Existence of inverse.

$\rightarrow$

Ex

$\Rightarrow$ ① $(\mathbb{Z}, +)$ ② $(\mathbb{Q}, \cdot)$ $\mathbb{Q} \neq 0$

or

A monoid $(A, *)$ is said to be group if each element of A has its inverse in A .

Ex ③ $(\mathbb{Z}^+, \cdot) \rightarrow$ monoid but not a group

b/c $2 \in \mathbb{Z}^+$ but its inverse $\frac{1}{2} \notin \mathbb{Z}^+$

Q: Show that the $G = \{2^n \mid n \in \mathbb{Z}\}$ is a group for multiplication

(RTU-2022)
(Mark-2)

Sol:

Abelian OR Commutative Group $\Rightarrow$

A group $(G, *)$ is said to be abelian or commutative if $a * b = b * a, \forall a, b \in G$.

Ex $\Rightarrow (\mathbb{Z}, +) \Rightarrow$ abelian group.

Q: Consider an algebraic system $(G, *)$, where G is the set of all non-zero real numbers and $*$ is a binary operation defined by $a * b = \frac{ab}{4}$, show that $(G, *)$ is an abelian group. (RTU-2024) (10-M)

Sol: Given $a * b = \frac{ab}{4} \quad a, b \in G \neq 0$ not

Closure $\Rightarrow$ let $a, b \in G$

$\Rightarrow a$ and b are non-zero real nos

$\Rightarrow ab$ is also a non-zero real no.

$$\Rightarrow \frac{ab}{4}$$

$$\Rightarrow \frac{ab}{4} \in G$$

$$\Rightarrow a * b \in G$$

$\Rightarrow G \rightarrow$ Closed.

Associative $\Rightarrow$ let $a, b, c \in G$

$$a * (b * c) = a * \left(\frac{bc}{4}\right) = \frac{a(bc)}{4 \cdot 4} = \frac{abc}{16} \quad \text{--- (1)}$$

$$\text{also } (a * b) * c = \left(\frac{ab}{4}\right) * c = \frac{\frac{ab}{4} \cdot c}{4} = \frac{abc}{16} \quad \text{--- (2)}$$

$$\therefore \text{from (1) \& (2)} \Rightarrow a * (b * c) = (a * b) * c$$

$$\Rightarrow \forall a, b, c \in G$$

$\Rightarrow G \rightarrow$ Associative

Existence of identity $\Rightarrow$

let $e \rightarrow$ identity element

$$a * e = a \quad \forall a \in G$$

$$\Rightarrow \frac{ae}{y} = a$$

$$\Rightarrow e = y \in G$$

$\therefore y$ is the identity element in G .

Existence of inverse $\Rightarrow$

let $b \rightarrow$ inverse of $a \in G$

$$\text{then } a * b = e$$

$$\frac{ab}{y} = e$$

$$\Rightarrow b = \frac{1b}{a} \in G \quad (\because a \neq 0)$$

Abelian $\Rightarrow$

let $a, b \in G$, then

$$a * b = \frac{ab}{y}$$

$$b * a = \frac{ba}{y}$$

$$= b * a$$

$$\Rightarrow a * b = b * a \quad \forall a, b \in G$$

Therefore $(G, *)$ is an abelian group.

Q: If G is a group such that $(ab)^2 = a^2b^2$ for all $a, b \in G$, then show that G must be an abelian. (RTU-2023)

sol: Given $\Rightarrow a^2(ab)^2 = a^2b^2$

$$\Rightarrow (ab)(ab) = (aa)(bb)$$

$$\Rightarrow \cancel{a}(ba)b = \cancel{a}(ab)b$$

(by left cancellation law)

$$\Rightarrow (ba)b = (ab)b$$

$$\Rightarrow ba = ab$$

(by right — —)

$$\forall a, b \in G$$

So, G is an abelian.

[NOTE $\Rightarrow$ Given $(ab)^2 = a^2b^2$, then $G \rightarrow$ abelian if $ab = ba$]

Composition Table for Groups $\Rightarrow$

Let $S = \{a_1, a_2, \dots, a_n\}$ is finite set, $*$ is a composition in S .

$*$	a_1	a_2	a_3	$\dots$	a_n
a_1	$a_1 * a_1$	$a_1 * a_2$	$a_1 * a_3$	$\dots$	$a_1 * a_n$
a_2	$a_2 * a_1$	$a_2 * a_2$	$a_2 * a_3$	$\dots$	$a_2 * a_n$
a_n	$a_n * a_1$	$a_n * a_2$	$a_n * a_3$	$\dots$	$a_n * a_n$

Eg $S = \{1, -1, i, -i\}$

Q: Show that the set $G = \{A, B, C, D\}$, where

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, D = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

is an abelian group for matrix multiplication.

Sol:

	A	B	C	D
A	A	B	C	D
B	B	A	D	C
C	C	D	A	B
D	D	C	B	A

Composite table

Closure $\Rightarrow$ All entries in the table are the element of G . so $G \rightarrow$ Closed.

Associative $\Rightarrow$

Since Matrix multiplication is always associative,
thus associativity holds in G .

$$\text{eg} \Rightarrow A \times (B \times C) \quad (A \times B) \times C$$

$$= A \times D$$

$$B \times C$$

$$= D$$

$$D \times D = D$$

Identity element $\Rightarrow$ The first row of the table,
headed by A , coincides with the top row. Therefore
 A is the identity element in G .

$$A \cdot e = A \Rightarrow e = A$$

$$B \cdot e = B \Rightarrow e = A$$

$$C \cdot e = C \Rightarrow e = A$$

$$D \cdot e = D \Rightarrow e = A$$

Existence of inverse $\Rightarrow$ From the table,

$$A \cdot A = A \Rightarrow A^{-1} = A$$

$$A = I \text{ (Unit matrix)}$$

$$B \cdot B = B \Rightarrow B^{-1} = B$$

$$C \cdot C = A \Rightarrow C^{-1} = C$$

$$D \cdot D = A \Rightarrow D^{-1} = D$$

So, each element of G has its inverse in G .

Hence, $G \rightarrow$ Group.

Now, The table is symmetrical about the leading diagonal.
So Composition is Commutative.

$\therefore G \rightarrow$ abelian group.

Subgroup $\Rightarrow$

A non-empty subset H of a group G is said to be a subgroup of G if the composition in G is also a composition in H and for this composition H itself a group.

Eg $\Rightarrow$ The multiplicative group $\{1, -1\}$ is a subgroup of the multiplicative group of the $\{1, -1, i, -i\}$.

NOTE $\Rightarrow$ ~~If H and K are two subgroups of a group G then HAK is also a subgroup of G .~~

NOTE $\Rightarrow$ To prove that any set H is a subgroup of a group G then we have to show that

- (i) H is a non-empty
- (ii) H is a subset of G .
- (iii) $a \in H, b \in H \Rightarrow ab^{-1} \in H$.

~~If~~ If $(G, *)$ be a group, and $H \subseteq G$ then $(H, *)$ be itself a group.
 H is a subgroup of G .

Q: If H and K are two subgroups of a group G then HAK is also a subgroup of G .

Sol: Given $\Rightarrow H \subseteq G, K \subseteq G$

Group Homomorphism $\Rightarrow$ (RTU-2024)

Let $(G, *)$ and $(G', *)$ $\rightarrow$ two groups

and $f: G \rightarrow G' \rightarrow$ a mapping

then f is called a group homomorphism if

for all $a, b \in G$

$$f(a * b) = f(a) *' f(b).$$

Q: Show that the mapping $f: (\mathbb{Z}, +) \rightarrow (\mathbb{Q}, \cdot)$ defined by $f(x) = e^x$ is a homomorphism.

Sol: let $x, y \in \mathbb{Z}$

$$\begin{aligned} f(x+y) &= e^{x+y} = e^x \cdot e^y \\ &= f(x) \cdot f(y) \end{aligned}$$

$$\left\{ \begin{aligned} f(x+y) &= f(x) *' f(y) \\ f(x+y) &= f(x) \cdot f(y) \\ e^{x+y} &= e^x \cdot e^y \end{aligned} \right.$$

Hence f is homomorphism.

Q: Show that the mapping $f: (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$ defined by $f(n) = n+1, n \in \mathbb{Z}$ is not a homomorphism.

Sol: let $x, y \in \mathbb{Z}$

$$f(x+y) = x+y+1 \quad \text{--- (1)}$$

$$\text{also } f(x)+f(y) = (x+1) + (y+1) = x+y+2 \quad \text{--- (2)}$$

from (1) & (2), $f(x+y) \neq f(x) + f(y)$

Hence $f \rightarrow$ not a homomorphism.

Group Isomorphism $\Rightarrow$

Let $f: (G, *) \rightarrow (G', \#)$ be a one-one mapping from a group G onto a group G' such that

$$f(a * b) = f(a) \# f(b) \quad \forall a, b \in G$$

then f is called group isomorphism from G to G'

or G is isomorphic to G' .

Symbolically, we write $G \cong G'$

Working Procedure to prove that G is isomorphic to G' $\Rightarrow$

(i) Define a mapping $f: G \rightarrow G'$

(ii) Show that f is one-one

(iii) show that f is onto

(iv) show that f is a homomorphism.

Q: Let G be a group. Show that the function $f: G \rightarrow G$ defined by $f(a) = a^{-1}$ is an isomorphism. ~~if~~ ~~and~~

Sol: Given mapping $f: G \rightarrow G$

one-one $\Rightarrow$ let $f(a) = f(b) \quad a, b \in G$

$$\Rightarrow a^{-1} = b^{-1}$$

$$\Rightarrow b = a$$

$$\Rightarrow f \rightarrow \text{one-one.}$$

$$\begin{aligned} \text{onto} \Rightarrow f(a) &= \bar{a}' & \begin{array}{c} a \rightarrow \bar{a}' \\ \bigcirc \quad \bigcirc \end{array} \\ f(a') &= (\bar{a}')' \\ f(a') &= a & \begin{array}{c} a \leftarrow \bar{a}' \\ \bigcirc \quad \bigcirc \end{array} \end{aligned}$$

$\Rightarrow$ each element of codomain has its preimage in domain.

~~Homom~~ $\Rightarrow f \rightarrow \text{onto}.$

Homomorphism $\Rightarrow$ Let $x, y \in G$ then

$$\begin{aligned} f(xy) &= (xy)' \\ &= x' y' \\ &= f(x) f(y) \end{aligned}$$

$$\begin{pmatrix} f(xy) \\ (xy)' \\ x' y' \end{pmatrix}$$

$$\begin{pmatrix} f(x) f(y) \\ x' y' \end{pmatrix}$$

$\Rightarrow f$ is homomorphism.

Hence, f is an isomorphism from G to G .

Permutation Group $\Rightarrow$ (R7V 2024)

A permutation of a finite set S is a bijection from set S to itself.

$$f: S \xrightarrow[\text{onto}]{\text{one-to-one}} S ; \text{ if } a \in S, \text{ then } f(a) \in S.$$

Notation $\Rightarrow$ let $S \rightarrow$ finite set of n elements.

$$S = \{a_1, a_2, \dots, a_n\}$$

$$\therefore F = \begin{pmatrix} a_1 & a_2 & \dots & a_n \\ f(a_1) & f(a_2) & \dots & f(a_n) \end{pmatrix}$$

Eg $\Rightarrow S = \{1, 2, 3, 4\}$ and $f: S \rightarrow S$

$$f(1) = 2, \quad f(2) = 3, \quad f(3) = 4, \quad f(4) = 1$$

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

COSETS $\Rightarrow$

Let H be a subgroup of a group G . and $a \in G$, then the set $aH = \{ah : h \in H\}$ is called left coset of H in G .

and $Ha = \{ha : h \in H\} \rightarrow$ right coset of H in G .

$$H * a = \{h * a : h \in H\}$$

Example $\Rightarrow$

① let $G = (\mathbb{Z}, +)$ and $H = 2\mathbb{Z} = \{0, \pm 2, \pm 4, \dots\}$

then

right coset of H in $G \Rightarrow$

$$H + 0 = \{0, \pm 2, \pm 4, \dots\} = H$$

$$H + 1 = \{0+1, \pm 2+1, \pm 4+1, \dots\} = H+1$$

$$H + 2 = \{0+2, \pm 2+2, \pm 4+2, \dots\} = H$$

$$H + 3 = \{0+3, \pm 2+3, \pm 4+3, \dots\} = H+1$$

$$H + 4 = \{0+4, \pm 2+4, \pm 4+4, \dots\} = H$$

$$H + 5 = \{0+5, \pm 2+5, \pm 4+5, \dots\} = H+1$$

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Hence H has only two distinct cosets $H, H+1$ in $\mathbb{Z}$.

②

Q: find all the cosets of $3\mathbb{Z}$ in the group $(\mathbb{Z}, +)$

a): let $H = 3\mathbb{Z} = \{\dots -6, -3, 0, 3, 6, \dots\}$

$$0 + H = H + 0 = \{\dots -6, -3, 0, 3, 6, \dots\} = H$$

$$1 + H = H + 1 = \{\dots -5, -2, 1, 4, 7, \dots\}$$

$$2 + H = H + 2 = \{ \dots, -4, -1, 2, 5, 8, \dots \}$$

$$3 + H = H + 3 = \{ \dots, -3, 0, 3, 6, 9, \dots \} = H$$

$$4 + H = H + 4 = \{ \dots, -2, 1, 4, 7, 10, \dots \} = H + 1$$

$$5 + H = H + 5 = \{ \dots, -1, 2, 5, 8, 11, \dots \} = H + 2$$

$$6 + H = H + 6 = \{ \dots, 0, 3, 6, 9, \dots \} = H + 0$$

:

:

Therefore there are three cosets of H in G

i.e. $H, H+1, H+2, \underline{H}$.

note $\Rightarrow$ There are n cosets of $n\mathbb{Z}$ in $(\mathbb{Z}, +)$.

$(+, \cdot)$ group with $n\mathbb{Z}$ for $n \neq 0$ with the binary